

PROPULSION SYSTEM INTERNSHIP

WEEK 1 : PLANNING

Engine Cycle Study : Design point analysis , Thrust calculation and Component sizing

DOCUMENT AND TRAINEE DETAILS

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Role	Propulsion System Intern
Reference engine	CFM56 -7B26 (Boeing 737NG Turbofan)
Purpose	Research,Scope definition,Method selection , Engine cycle study
Duration	4 weeks (28 th September 2026 - 26 th October 2026)



Fig 1



fig2

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1. OVERVIEW

In a span of four weeks of the Propulsion system internship, this document explains how each work and sections will be carried out and completed. The study dives in High Bypass, two spool, separate flow civil Turbofan and its design point. Calculating the net thrust , Thrust Specific Fuel Consumption TSFC , sizing the main flow from the values obtained and finally validating every data against the available published data for a reference engine.

This week is a base point for research and it shows that an engineer follows a sequence of steps when dealing with propulsion systems. The sequence is as :- defining the operating point and design choices , station by station modelling the engine , verifying the mathematics and the hand calculation hence these results are finally checked in certified data . This is the same model I have employed for this internship.

ITEMS	EXPLANATION
Problem	Design point cycle analysis and sizing of a Civil turbofan
Reference engine	CFM56-7B26 turbofan
Design point	Sea level static takeoff - primary , cruise at 35000 ft, Mach 0.8 - as time permits
Methods	Station by station 1D thermodynamics model following Mattingly and Saravanamutto, SAE AS755 station numbering
Tools	Python (Numpy , Matplotlib), pyCycle , Github
Success justification	Untuned thrust reported, then calibrated value within 1% ,TSFC within +/- 10% and fan diameter within +/- 5% . Each result is checked carefully
Outputs	Everything falls in week 2 for thrust calculation , week 3 sizing and validation and then week 4 for a full dossier and a presentation.

2. ROLE AND PROBLEM OVERVIEW

Each engineer in propulsion system makes sure the engine delivers thrust it needs at a very reasonable amount of fuel by considering fundamental factors such as temperature, pressure and the size limits without forgetting the hardware.

It is compulsory for an engineer to run a cycle study which is a thermodynamics baseline which shows how the engine will burn fuel and performance of the engine. These results determine decision on the size of components,the material required and if it meets aircraft requirements.

I am basically mimicking building a small scale but a version of that engineering craftsmanship workflow and the statement problem that I will be working on will be

“ Using station by station point model, calculate the sea level static takeoff thrust and TSFC of the CFM56-7B26 (ISA and mach 0),estimation of the fan, core and nozzle flow areas and comparing with ICAO and EASA data published “

How is this work fit into 4 weeks plan?

WEEK	TASK	ROLE
1	Planning and research	In this document, research,scope of the engine study,references and methods .
2	Thrust calculation	Operational and working point model with all the inputs and results
3	Component sizing and validation	Sizing from the second week fruits and comparing everything with the available public data also analyzing errors and revisions
4	Final report and summary of the whole internship	Report and showing how each week's output has been contributing to the following week

3.RESEARCH ON HOW ENGINEERS HANDLE ENGINE CYCLE STUDY

Engineers have been approaching and handle engine study in vast ways and these are the main and fundamental ways to deal with .

(a) Cycle analysis

This is also categorized in different types of engine cycle types as follows

Type of analysis	Function of the analysis	Relationship to my project
Design point	This calculates performance in a certain condition such as Overall Pressure ratio and Bypass Pressure Ratio	It is the fundamental part on my intern especially in week 2 and 3
Parametric design	It checks the design to show alignments in thrust and thrust specific fuel consumption	It gives direction and it is optional
Off design	This finds performance in other secondary factors such as different mach number or altitude change.	This can be noted and used as a future work or if time permits it will be done
Ideal cycle	This assumes there are no losses in our system	This is used to study the how the system behaves under ideal thermodynamic conditions.
Real cycle	This is the basis for real life engine operation as it comprises of losses, efficiencies pressure ratio etc	This is fundamental as it shows how the system deviated from the ideal cycle

Brayton cycle gives out fundamentals such as high pressure ratio and turbine temperature improve thermal efficiency also higher bypass ratio gives out higher propulsive efficiency. Real engines show deviation because they are not 100% efficient there are some losses in the inlet , combustion , nozzles and also mechanical losses not forgetting properties change due to temperature change and the air fuel ratio.

I will model a real ideal cycle then I will add some efficiencies and some losses to see how data have been deviating overtime.

(b) Station by Station method

Engine is primarily the chain of components and processes. In this whole internship, the numbering system used is that of SAE AS755 which is the successor of the ARP755 .

For a two spool separate flow turbofan are as follows

Station	Location	Characteristics
0	Free stream	T_0 , p_0 , V_0 , Mach number
2	Fan face (inlet exit)	T_{t2} , p_{t2} after inlet
13	Fan exit , bypass stream	T_{t13} , p_{t13} ,bypass mass flow
21/25	Fan exit ,core / High Pressure Combustion inlet HPC	T_{t25} , p_{t25}
3	HPC exit / combustor inlet	T_{t3} , p_{t3} ,compressor work
4	Combustor exit , HPT inlet	T_{t4} (TIT) , p_{t4} , fuel air ratio
45	HPT exit / LPT inlet	T_{t45} , p_{t45}
5	LP exit	T_{t5} , p_{t5}
8/9	Core nozzle throat / exit	Core escape velocity V_9 and pressure P_9
18/19	Bypass nozzle throat / exit	Bypass exit velocity V_{19} , pressure p_{19}

(c) Industry approach

Corporations such as GE,MiG which manufacture engines use advanced software such as NPPS and GasTurb . However there have been some alternative of which I have employed from open source entities such as NASA pyCycle . From this what I learn is I have to check mathematics and equations first also each input should be traceable throughout the process.

4. STANDARDS AND INDUSTRY REFERENCES

Consider the table below of the standards and the reason why the specific standard is used .

Standard	Reason
Mattingly , Elements of propulsion - Gas Turbines and Rockets	This is for the working design point equations for turbofans
Saravanamuttoo et al , Gas Turbine theory	Real cycle effects and components efficiency
Walsh and Fletcher , Gas Turbine performance	For real industry style calculations and performance and uncertainties
SAE AS755	This is for engine station number as they are categorized and identified by numbers
SAE ARP1420	Guiding in how distortion might occur and goes beyond the 1D scope
ICAO Aircraft Engine emissions databank	Identifying the takeoff fuel flow that is used to determine Thrust Specific Fuel Consumption as a target for validation
ISO 2533 for ISA	Reference for ambient temperature and pressure at altitudes
EASA type certificate data sheet and manufacturer datasheet	This is for verification on published values such as Thrust , Bypass Pressure Ratio BPR , Overall Pressure Ratio OPR and also identifying limits

5. SCOPE , ASSUMPTIONS AND CONSTRAINTS

In Scope

- Design point performance of a two spool , separate flow high bypass turbofan
- Calculation of net thrust , TSFC , thermal efficiency , propulsive efficiency and overall efficiency
- Preliminary fan design , compressor , turbine and nozzle flow areas and diameters
- Comparison with published data and documenting errors for later analysis

Out of Scope

- Off design maps and transient behaviour
- 3D CFD , structural , thermal or mechanical detailed design
- Emissions , noise and installation effects
- Proprietary engine data

Reference Engine and Design point

In this study the reference engine is the CFM56-&B26 which is a two spool , separate flow , high bypass turbofan used on the Boeing 737 Next Generation. The reason for this choice is because the engine itself is certified , a lot of public data materials for verification. According to sources this engine has 61 in diameter fan , behind the fan there is a 3 stage low pressure compressor LPC followed by nine stage High Pressure Compressor HPC and one stage high pressure turbine HPT to drive the fan and lower pressure compressor.

The engine chosen is pre Tech 56 with maximum takeoff thrust of 27300 lbs . It is shown that takeoff thrust is independent of the ambient temperature and the flat temperature also the continuous thrust is higher than the takeoff thrust which is also independent of ambient temperature.

All these values are confirmed through EASA certification . In this document the design point level is sea level static takeoff (mach 0 , 0m ,ISA) and cruising altitude at 35000 feet and mach number 0.8.

ASSUMPTIONS

Number	Assumption	Reason
A1	1D steady flow at each engine station	This is for ideal preliminary cycle analysis
A2	Normal atmosphere ISA as there is no wind or humidity	Certifications are provided while considering ISA
A3	Constant specific heats initially and then for cold and hot section follows	We need to compute for simple check and then where there is heat change
A4	Values of efficiency of components and pressure losses from the textbook	Sometimes public data maybe altered by some sources
A5	Jet A fuel, LHV about 43 MJ/kg, fixed combustorefficiency	Identity for civil aviation fuel for jet engines
A6	Cooling air and bleed are treated as simple fraction or ignored initially	Detailed cooling data are proprietary, the effect is examined in the error analysis
A7	Presence of separate flow nozzles with a heck if it is choked	This is because chocking changes the pressure in the nozzle

CONSTRAINTS

- Turbine inlet temperature is kept within a realistic limit for the CFM56-7B26
- Pressure ratios are kept within range of the CFM56-7B26 engine
- Only publicly available data are used
- In a span of four weeks cruise condition and the parametric work are optional

INPUT VALUES AND THEIR SOURCES

Each value is marked published or assumed or taken from a text book stating points is recorded in week 2 and 3

INPUT	VALUE	SOURCE
Take off thrust	26300 lbf approx 117kN	EASA datasheet , FADEC
Bypass ratio	5.1 : 1	Edb emmision databank from ICAO , EASA
Overall pressure ratio	27.2 for takeoff and 32.7 / 32.8 for cruising	Wikipedia , EASA , ICAO emmision databank
Fan tip diameter	61.0 inch	
Core layout	Fan , booster ,9 stage HPC , 1 stage HPT , 4 stage LPT	Published in trade press and aircraft operators and owners guide
Takeoff airflow	341 kg/s	Aerocontact : under ISA 752 lbs/sec
Takeoff fuel flow	1..221 kg/s	ICAO
Flat rating temperature	30 deg celcius	EASA : ISA+15 and ISA is 15 deg
Fan pressure ratio	1.65	textbook
Turbine inlet temperature Tt4	1280 - 1300 deg celcius	IISTR journals : advanced exergy analysis on the turbofan engine
Component efficiencies and pressure losses	To be obtained each at a specific time	Assumed from text book
Fuel lower heating value	43.15 MJ/kg	BP aviation , Wikipedia , Bombardier

6. METHODOLOGY

Calculation sequence

1. **Atmosphere:** To obtain T_0 , p_0 from ISA at the chosen altitude
2. **Inlet:** applying total temperature and pressure ratios for ram compression and inlet recovery to get T_{t2} and p_{t2} .
3. **Fan and compressors:** use pressure ratio and efficiency to get exit temperature and specific work.
4. **Combustor:** with T_{t4} fixed, calculate fuel-air ratio f and fuel flow.
5. **Turbines:** find turbine temperature drop from the power balance with the fan and compressor.
6. **Nozzles:** check choking, then compute exit velocity and pressure for core and bypass streams.
7. **Performance:** sum the thrust terms and compute TSFC and efficiencies.
8. **Sizing:** use continuity at chosen Mach numbers to find areas and diameters.

Key equations

Flight speed and ram conditions:

$$V_0 = M_0 \sqrt{\gamma R T_0} \quad ; \quad T_{t0} = T_0 [1 + (\gamma - 1)/2 \cdot M_0^2] \quad ; \quad p_{t0} = p_0 [T_{t0}/T_0]^{\gamma/(\gamma - 1)}$$

Fan or compressor temperature ratio from pressure ratio and isentropic efficiency

$$T_{c} = T_{t,out}/T_{t,in} = 1 + (\pi_c^{\gamma/(\gamma - 1)} - 1)/\eta_c$$

Fuel air ratio from combustor energy balance

$$f = (c_{p,t} \cdot T_4 - c_{p,c} \cdot T_3) / (\eta_{b,h} \cdot h_{PR} - c_{p,t} \cdot T_{t4})$$

Turbine power balance

$$\eta_m \cdot (1 + f) \cdot c_{p,t} \cdot (T_{t4} - T_{t5}) = c_{p,c} \cdot (T_{t3} - T_{t2})$$

Nozzle exit velocity for the expanded stream

$$V_9 = \sqrt{2 c_{p,t} \cdot T_{t9} [1 - (p_9/p_{t9})^{\gamma/(\gamma - 1)}]}$$

Net thrust of separated flow including pressure

$$F = \dot{m}_{core} [(1 + f)V_9 - V_0] + \dot{m}_{bp} (V_{19} - V_0) + A_9 (p_9 - p_0) + A_{19} (p_{19} - p_0)$$

Performance measures

$$TSFC = \dot{m}_{core} / F \quad \Delta KE = \frac{1}{2} [\dot{m}_{core} (1 + f) V_9^2 - \dot{m}_{bp} V_{19}^2 - \dot{m}_0 V_0^2]$$

$$\eta_p = F \cdot V_0 / \Delta KE \quad \eta_{th} = \Delta KE / (\dot{m}_{fuel} \cdot h_{PR}) \quad \eta_o = \eta_{th} \cdot \eta_p$$

7. VERIFICATION AND VALIDATION

Verification asks whether I solved the equations correctly. Validation asks whether the equations represent the real engine. In this whole internship I will do both in depth.

Consider the following table

Event	Pass criteria
Fabricating a textbook turbofan example	Matching the book results within rounding
Limiting case: BPR tends to 0 (turbojet)	Results approach turbojet behaviour
Limiting case: zero flight speed	Thrust is finite and only from the jet terms
Power balance: turbine work equals compressor and fan work	Residual below 0.1 %
Unit and dimension check on every equation	No unit inconsistencies

Validation targets

Quantity	Target	Source of comparison
Takeoff net thrust	Within +/- 10%	Data sheet
Airflow and BPR	Matching input within 1%	Data sheet
Overall pressure ratio	Matching input	Data sheet
TSFC	Within +/- 10%	Published data
Fan tip diameter	Within +/- 5%	Data sheet

Uncertainty and error sources

In Week 3 I will change each of the main uncertain inputs (efficiencies, $Tt4$, pressure losses, cooling air fraction, gas properties) by a plausible amount, one at a time, and record the change in thrust and TSFC. This shows which assumptions matter most and how much the conclusions depend on them.

8 TOOLS

Tool	Reason
Python with NumPy and Matplotlib	Free, transparent and repeatable. Every equation is visible in the code, and results and plots can be regenerated by anytime
GasTurb or NPSS / pyCycle	Industry-standard tools that provide an independent check of the model. If unavailable, textbook examples take their place.
GitHubG	Version control and public evidence of the work; required for the technical internships.
Microsoft word	Formation for report

9. DELIVERABLES , REPOSITORY AND DOCUMENTATION STANDARDS

Repository structure

README.md	:	project summary , how to run the code , and links to documents
docs/	:	weekly report and the final short presentation summary
src/	:	python code for the cycle model and calculation
data/	:	reference engine data and sources
results/	:	graphs and tables generated from code
tests/	:	verification checks

Documentation standards

- SI units throughout, with conversions from imperial data shown
- Every result shown with its inputs, method and source.
- Consistent significant figures, with rounding only in the final presentation of results.
- Every assumption numbered and referenced when used.
- Sources cited for all published data.
- A revision log recording each change to the model and the reason.

10. RISKS AND MITIGATION

Risk	Likelihood	Effect	mitigation
Limited public data for the chosen engine	Medium	High	Using data sheets initially
Results differs from published data by more than 10%	high	Medium	Checking units and station definition and reviewing efficiencies
Coding or unit errors	Medium	High	Hand checking the ideal cycle and testing the limiting cases
Time runs out	Medium	High	Prioritize takeoff design first

11. ABBREVIATIONS

BPR	:	Bypass pressure ratio
EASA	:	European Union Aviation Safety Board
HPC	:	High pressure compressor
HPT	:	High pressure turbine
ISA	:	International Standard Atmosphere
LPT	:	lower pressure turbine
OPR	:	Overall pressure ratio
TSFC	:	Thrust specific fuel consumption

12. Reference List

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- SAE International. ARP755: Gas Turbine Engine Performance Station Identification and Nomenclature (current revision).
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- Hendricks, E. S. and Gray, J. S. (2019). pyCycle: A tool for efficient optimization of gas turbine engine cycles. Aerospace, 6(8), 87.
- EASA type-certificate data sheet and manufacturer datasheet for the chosen reference engine
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- aircraft operators and owners guide
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